Frozen Strait (Arctic / Naval / Naval + Air + Ice Operations)

Executive Summary

Frozen Strait is a **hybrid Arctic naval battlefield** built for **naval warfare, air operations, ice-field maneuvering, and autonomous multi-domain coordination**. The terrain features fractured sea ice, narrow straits, iceberg clusters, frozen channels, coastal cliffs, submarine vents, and forward Arctic stations. The map emphasizes Obsidian Protocol's autonomous naval routing, air-sea coordination, sensor-fusion under Arctic interference, and logistics under unstable ice conditions. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km^2 of Arctic maritime terrain. Key features include ice-breaker routes, naval chokepoints, airspace corridors, submarine vents, and multi-domain combat across sea, air, and ice.

Design Goals & Gameplay

Naval Warfare in Ice-Choked Waters
Frozen Strait prioritizes ship-to-ship combat, torpedo lanes, ice-breaker routes, and hull-hazard navigation. Ice floes create dynamic cover and movement obstacles.

Air Operations & Reconnaissance
VTOL aircraft, drones, and long-range recon planes operate above the strait. Airspace corridors between cliffs and icebergs define movement and targeting.

Ice-Field Maneuvering
Ground units traverse frozen channels, ice ridges, and unstable floes. Heavy armor risks breaking ice; light units excel in mobility.

Autonomy & Multi-Domain Coordination
Autonomous units coordinate naval, air, and ice-field forces simultaneously. Units adapt to shifting ice patterns, fog pockets, and sonar interference.

Sensor & Information Warfare
Sea fog, ice mist, thermal distortion, sonar scatter, and magnetic interference degrade sensors. Relay towers on cliffs restore clarity. Destroying relays creates blind pockets across the strait.

Logistics & Resources
Fuel caches, coastal stations, and submarine vents serve as resupply nodes. Convoys must navigate ice-breaker routes and avoid ambush-prone channels.

Deployment & Progress
A fixed deployment budget (~10,000 points) forces players to choose between naval dominance, air superiority, or ice-field mobility. Blue deploys from the southwest coastal station; Red deploys from the northeast cliffside command.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling
10×10 m physical area ≈ 1 km^2 Arctic maritime zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations
Blue HQ (Coastal Station Bravo) — (0,0), 150×150 m Docks, fuel tanks, drone pads, Arctic staging.
Red HQ (Cliffside Command Spire) — (1000,1000), 150×150 m Heated bunker with radar mast, AA emplacements, and sonar arrays.
Central Frozen Strait — (500,450), 350×250 m Fractured ice, open water lanes, iceberg clusters, naval combat zone.
Submarine Vent Field — (250,800), 120×120 m Thermal vents, sonar distortion, underwater hazards.
Ice Ridge Network — (800,300), 200×200 m Wind-carved ridges, frozen tunnels, flanking routes.
Ice-Breaker Channel — (0,450 → 400,900) Primary naval chokepoint: narrow, hazardous, constantly shifting.
Frozen Shore Route — (300,0 → 450,350) Safer ground route with partial cover.
Cliff Relay A — (650,150), +240 m Long-range sensor vantage point.
Cliff Relay B — (150,650), +260 m Comms relay hub and sonar control node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Ice-Breaker Channel
Major naval chokepoint; controlling it determines sea movement.

Submarine Vent Field
Thermal distortion disrupts sonar; ideal for submarine ambushes.

Frozen Strait Center
Long-range naval and air engagements; dynamic ice cover.

Ice Ridge Network
Interior flanking routes; unpredictable cover due to drifting snow and ice.

Sightlines
Cliff Relay A provides long-range sensor coverage (~350 m). Sea fog and ice mist create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW coastal station + naval staging.
Red Deployment: NE cliffside command spire + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain
Ice floes, frozen channels, cliffs, submarine vents, coastal stations.

Vegetation
None — Arctic maritime zone.

Buildings & Props
Military: AA guns, radar dishes, sonar arrays, bunkers. Industrial: Generators, pipelines, docks. Logistics: Ships, cargo crates, fuel drums, drones.

Effects
Sea fog, ice mist, snowstorms, thermal distortion, wave spray.

Audio
Wind, cracking ice, ship engines, sonar pings, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Ice/Terrai	3	200k	600k
Naval/Inc	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget
2048^2 for terrain/buildings, 1024^2 for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors
Single spatial anchor for entire map.

Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer
Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototype Layout — Month 1 Block-out strait, ice ridges, submarine vents, AR anchor test.
2. Core Systems — Month 2 NavMesh (ice + land), naval autonomy, air-sea coordination.
3. AR Integration — Month 3 Anchors, plane detection, basic interactivity.
4. Content Fill — Month 4–5 Final models, textures, lighting, audio.
5. Gameplay & Polish — Month 6–7 Objectives, balance, VFX (sea fog, ice mist, thermal distortion).
6. Testing & QA — Month 8 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy
Verify naval routing, air-sea coordination, ice-field navigation.

Comms/Sensors
Test relay destruction, fog-of-war updates, sonar interference.

Performance
≥60 Hz AR render on target device.

QA Checklist
Navigation, naval logic, deployment budget enforcement, anchor drift, sensor accuracy.